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Measurement of static pressure profiles of vortex flashing R134a flow expanded through convergent–divergent nozzles

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ABSTRACT

Vortex control is a novel two-phase convergent–divergent nozzle restrictiveness control mechanism which achieves flow control by adjustable nozzle inlet vortex strength. It can potentially provide flow control with less sacrifice of nozzle efficiency, which is important for two-phase ejector cooling cycle performance. It is also less vulnerable to clogging. However, the underlying mechanism behind vortex control is still unclear. In this study, static pressure profiles of vortex flashing R134a flow expanded through nozzles under various conditions have been measured. A 1D model for the estimation of vapor qualities in the initially subcooled flashing nozzle flow based on the measurement results was also developed. It was found that after the introduction of inlet vortex to the initially subcooled flashing nozzle flow, the pressure drop across the divergent part has been increased, which is caused by the increased vapor generation in the divergent part. The elevated nozzle throat pressure results in the nozzle behaving like being more restrictive. When the inlet vortex is applied to flashing nozzle flow with two-phase at the nozzle inlet or single-phase liquid nozzle flow, the influence of inlet vortex on the nozzle restrictiveness and the nozzle pressure profile is insignificant. The nozzle isentropic efficiency can be significantly increased by applying inlet vortex, which could be beneficial to ejector performance when the vortex nozzle is used in ejectors. In order to achieve satisfactory vortex control range and high nozzle isentropic efficiency, the vortex nozzle divergent part length as well as the divergent angle need to be appropriately sized.

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Mesurage des profils de pression statique de la vaporisation instantanée de R134a en vortex détendue à travers des tuyères convergentes-divergentes

Mots-clés: Tuyère; Frigorigène diphasique; Régulation de l'écoulement; Écoulement en vortex; Profil de pression; Éjecteur

1. Introduction

Refrigerant flow control with expansion devices is crucial to achieve optimal performance of air conditioning and refrigeration systems under varying working conditions. The most commonly used expansion devices that are capable of flow modulation include thermostatic expansion valves (TXV) and electronic expansion valves (EEV), which achieve flow control by changing expansion valve opening.

In vapor compression cycles using two-phase ejectors, refrigerant flow control can also provide significant improvement to cycle performance. Ejector cycle performance is sensitive to working condition changes. Different working conditions require different ejector geometries to achieve optimal performance. Slightly different geometries may result in substantially different COPs under the same conditions. In order to optimize ejector cycle performance under different working conditions and make ejector technology more suitable for actual applications, it is desirable to introduce an adjustable feature to the ejector. The ejector motive nozzle throat diameter (motive nozzle restrictiveness) is found to be one of the key dimensions that affect ejector cycle performance (Sarkar, 2012). It has a direct impact on motive mass flow rate.

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Nomenclature

Symbols

A	area [m^2]
D	diameter [m]
f	friction factor [dimensionless]
F	force [N]
$\dot{m}$	mass flow rate [g s^{-1}]
P	pressure [kPa]
Re	Reynolds number [dimensionless]
T	temperature [°C]
TNE	thermodynamic non-equilibrium [dimensionless]
V	velocity [m s^{-1}]
VS	vortex strength [dimensionless]
x	vapor quality [dimensionless]

Greek symbols

ϵ	surface roughness [m]
η	efficiency [dimensionless]
μ	viscosity [Pa s]
ρ	density [kg m^{-3}]
τ	shear stress [Pa]

Subscripts

axial	axial inlet
e	thermodynamic equilibrium
in	inlet
isentropic	isentropic
l	liquid
out	outlet
tangential	tangential inlet
total	total mass flow rate through the nozzle
tp	two-phase
v	vapor
w	wall

Previous methods to adjust the effective hydraulic motive nozzle throat diameter in order to optimize ejector cycle performance according to the working conditions include using adjustable needle, using expansion valve upstream of or in parallel with the motive nozzle, and using multiple ejectors in parallel.

For motive flow control with adjustable needle, reduction in work recovery efficiency might be caused as the needle moves towards the throat because of the higher frictional losses resulting from the additional surface area introduced by the use of needle and the increased flow speed near the throat, as well as the higher level of thermodynamic non-equilibrium. Elbel and Hrnjak (2008) were the first researchers to publish experimental results of introducing a variable two-phase ejector to a transcritical R744 system by installing a needle in the motive nozzle for high-side pressure control. By moving the needle towards the motive nozzle throat, gas cooler exit pressure has been increased from 9.70 MPa to 10.57 MPa, while cycle cooling capacity and COP have been increased from 4.7 kW to 5.1 kW and from 1.12 to 1.17, respectively. Meanwhile, ejector efficiency reduced by 43.1%. Using the needle to reduce the motive nozzle throat area increased the internal losses in the ejector. The benefits of high-side pressure control offset the losses in nozzle and ejector efficiencies.

For motive flow control with expansion valve upstream of or in parallel with the motive nozzle, irreversibilities are incurred during the isenthalpic expansion in the expansion valve as a cost of flow control, thus work recovery efficiency is reduced. Lawrence and Elbel (2019) experimentally investigated and compared two different methods (ejector with adjustable needle and ejector with expansion valve upstream of the motive nozzle) for controlling high-side pressure of transcritical R744 cycle. Both methods were found to be effective at controlling high-side pressure, but both methods also resulted in significant losses caused by the control mechanism. As high-side pressure is increased, there is a slight deviation in COP when comparing the two control methods, with the expansion valve-controlled cycle having up to 1.5% lower COP as high-side pressure is increased by up to 800 kPa. However, the maximum COP, achieved by increasing the high-side pressure by 200 kPa, is very similar for the two control methods. A small but noticeable difference in total work recovery efficiency between the two control methods can also be observed. As high-side pressure increases, the expansion valve control yields up to 15.3% lower work recovery efficiency.

The use of multiple ejectors in parallel for motive flow control has been successfully applied to supermarket refrigeration (Hafner et al., 2014). The advantage of parallel ejector arrangements is that the overall motive nozzle restrictiveness can be varied without a loss in efficiency due to the control mechanism, although the variation of overall motive nozzle restrictiveness is not continuous. In addition, the geometry of the mixing sections and the suction nozzles can also be optimized for different conditions. The major disadvantage of parallel ejector arrangement is the relatively higher cost compared to the other control mechanisms due to the use of multiple ejectors and valves.

A variable ejector with adjustable needle or expansion valve upstream of or in parallel with the motive nozzle can optimize ejector cycle performance under different conditions, but the work recovery efficiency is reduced as a cost of flow control. Meanwhile, the relatively higher cost of using multiple ejectors in parallel might limit its application to smaller systems. This provides motivation to develop alternative technology to control the motive nozzle restrictiveness.

A novel two-phase nozzle restrictiveness control mechanism called vortex control was proposed and characterized by Zhu and Elbel (2016). Vortex with adjustable strength is generated at the nozzle inlet for the control of flashing nozzle flow. Zhu and Elbel's experimental investigation on the influence of nozzle inlet vortex on the nozzle restrictiveness on initially subcooled flashing R134a flow expanded through convergent–divergent nozzles showed that the inlet vortex strength can change the restrictiveness of the two-phase convergent–divergent nozzle. The nozzle becomes more restrictive as the vortex strength increases. The influence of nozzle geometric parameters on the vortex nozzle control has been presented in Zhu and Elbel (2017) and the influence of operating conditions on the vortex nozzle control were presented in Zhu and Elbel (2018). With vortex control, the choked mass flow rate can be reduced by 42% when the refrigerant is R134a, P_{in} = 1030 kPa, T_{in} = 40.0 °C. The choked mass flow rate variation represents the largest mass flow rate control range that can be achieved by vortex control for certain vortex nozzle design and nozzle inlet conditions. The control range of inlet pressures and mass flow rates that can be achieved by vortex control appears to be large enough to be suitable for numerous technical applications. This novel mechanism, when applied to an ejector, can potentially provide flow control with less sacrifice of work recovery efficiency than with upstream expansion valve and internal needle. The isenthalpic expansion in the expansion valve-controlled ejector and the additional frictional losses near the throat in the ejector with adjustable needle can be avoided. The applied vortex is also believed to reduce the thermodynamic non-equilibrium of the nozzle outflow, which might be beneficial to ejector work recovery efficiency. Nozzle with vortex control is also less vulnerable to clogging since the nozzle flow rate can be reduced without reducing the flow cross-sectional area. A variety of different expansion devices, in addition to ejectors, could potentially benefit from this new control mechanism, including actively controlled flow metering devices for superheat

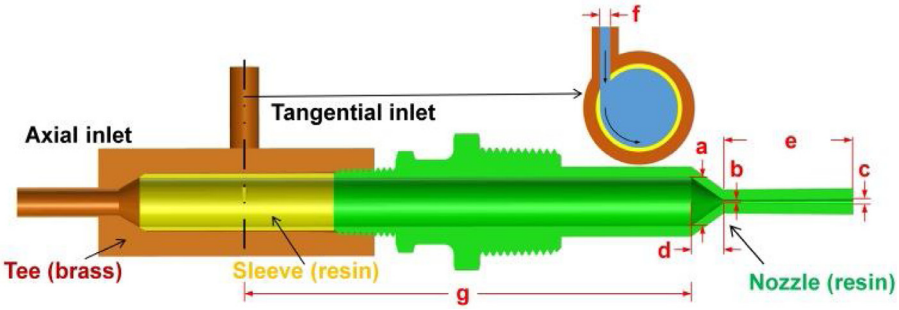


Fig. 1. Vortex nozzle composed of tee, sleeve and convergent–divergent nozzle.

control in subcritical applications or high-side pressure control in transcritical systems.

Little knowledge is available regarding the working principles of vortex control effect on flashing nozzle flow. In this study, measurement of the static pressure profiles of the vortex flashing R134a flows in convergent–divergent nozzles under different conditions was conducted, which can provide more insights into the vortex flashing nozzle flows and substantial contribution to the explanation of the vortex control effect discovered and characterized in previous publications. In the following sections, the test facility for measurement of the pressure profiles of the vortex flashing flow will be introduced first. The nozzle pressure profiles under different inlet (subcooled liquid or two-phase; with or without inlet vortex) and outlet conditions will be presented and discussed. A 1D model for the estimation of vapor qualities in the initially subcooled flashing nozzle flow based on the measurement results has been developed and the estimation results will be presented and discussed. Comparisons of the measurement and estimation results of nozzles with different geometric parameters will be shown. The new insights provided by the pressure profile measurements are essential to understand the working principles of vortex control effect.

2. Experimental apparatus and procedures

Several nozzles with controllable vortices at the nozzle inlet have been designed and manufactured for experiments, as shown in Fig. 1. The vortex nozzle is composed of three components: a T-shaped part made of brass, a sleeve and a nozzle, both made of an optically clear resin and manufactured with a Stereo Lithography Apparatus (SLA). SLA is one of the most widely used 3D-printing technologies. The tee-shaped part serves as the vortex generator. The tangential inlet on the tee allows flow to be injected tangentially and to mix with the axial flow, thus creating a vortex. The tee and the nozzle are joined by a conical thread and sealed by epoxy adhesive. The other conical thread on the nozzle is for connection with a visualization chamber. The sleeve is designed to provide a smooth transition between the tee part and the nozzle. The gap between the sleeve and the nozzle is minimized to reduce any additional disturbance introduced to the flow. The inner diameter of the sleeve is the same as that of the nozzle entrance. There is a tangential inlet on the sleeve. The tangential inlet on the tee and the tangential inlet on the sleeve are coaxial and have the same inner diameter. For visualization purpose, the flow needs to travel a distance from the tangential inlet to the starting point of the convergent part of the nozzle. This distance is called vortex decay distance, as shown in Fig. 1 with the letter 'g'. Because of the fluid viscosity and turbulence, the vortex strength will decay over this distance. Important dimensions of all the tested vortex nozzle have been summarized in Table 1 and these dimensions are shown with corresponding letters in Fig. 1.

The transparent convergent–divergent nozzle with pressure measurement channels and the cross-sectional view of the nozzle are presented in Fig. 2. There are six pressure measurement channels in total for each nozzle and they are numbered from 2 to 7 as shown in Fig. 2(b). Pressures at six different axial locations of each nozzle are measured independently through the pressure measurement channels. Only two of the six channels are shown in the cross-sectional view in Fig. 2(b). The other four channels are in different planes such that there is no conflict between channels. The channels are embedded in the 3D-printed design, so that channels did not have to be drilled separately. The tips of these channels in contact with the nozzle flow are of square shape with edge length of approximately 0.25 mm. This is the minimum channel tip size that can be 3D-printed without any clogging. Numbers 1 and 8 represent the nozzle inlet and outlet pressures, respectively. The axial locations of these measurement channel tips relative to the nozzle throat are presented in Table 2.

The layout of the experimental facility for measurement of pressure profiles of vortex flashing flows in convergent–divergent nozzles is shown in Fig. 3. A pumped-refrigerant-loop was used for adjustment of nozzle test conditions. The working fluid was refrigerant R134a. A visualization chamber was built from clear PVC pipe. The temperature readings were all obtained from ungrounded Type-T immersion thermocouples. The measured temperatures were regarded as total temperatures. Absolute pressures were read by piezo-electric pressure transducers. Pressures and temperatures in the upstream of the nozzle axial and tangential inlet control valves were measured. Pressures downstream of the control valves were also measured. For tests with two-phase flow at the nozzle inlet, the fluid upstream of the inlet control valves was single-phase liquid, while downstream of the control valves the fluid became two-phase after the expansion. The nozzle inlet vapor quality was calculated by the measured pressure and temperature of the single-phase liquid upstream of the valves and the pressure downstream of the valves and assuming isenthalpic expansion. The existence of two-phase at the nozzle inlet was confirmed by visualization through the transparent body of the nozzle. The axial inlet pressure was slightly lower than the tangential inlet pressure. This pressure difference was caused as a result of the refrigerant flowing through the tangential inlet with relatively small inner diameter (2.0 mm). The differences between the vortex nozzle axial inlet pressures and tangential inlet pressures were within 10 kPa when the tangential inlet mass flow rate was less than 13.0 g s^{-1} (single-phase liquid). The pressure difference increased to 28 kPa when the tangential inlet mass flow rate was 20.5 g s^{-1} (single-phase liquid). The axial inlet pressure was assumed to be the nozzle inlet pressure P_{in}. The pressure at different axial locations of the nozzle was measured by the same pressure transducer and control valves were used to switch between different pressure measurement channels on the nozzle. The nozzle outlet P_{out} was measured as well. Small mass flow rate through the axial or tangential inlet that leads to large temperature

Table 1
Important dimensions of tested vortex nozzles (each dimension is shown with corresponding letters in Fig. 1).

Nozzle #	1	2	3	4	5
(a) Nozzle inlet diameter (mm)	15.0	15.0	15.0	15.0	15.0
(b) Nozzle throat diameter (mm)	1.02	1.00	0.83	1.00	1.00
(c) Nozzle outlet diameter (mm)	1.37	1.72	1.19	1.18	1.71
(d) Nozzle convergent part length (mm)	9.9	9.9	9.9	9.9	9.9
(e) Nozzle divergent part length (mm)	20.0	40.0	20.0	20.0	20.0
(f) Tangential inlet inner diameter (mm)	2.0	2.0	2.0	2.0	2.0
(g) Vortex decay distance (mm)	168.0	168.0	168.0	168.0	168.0

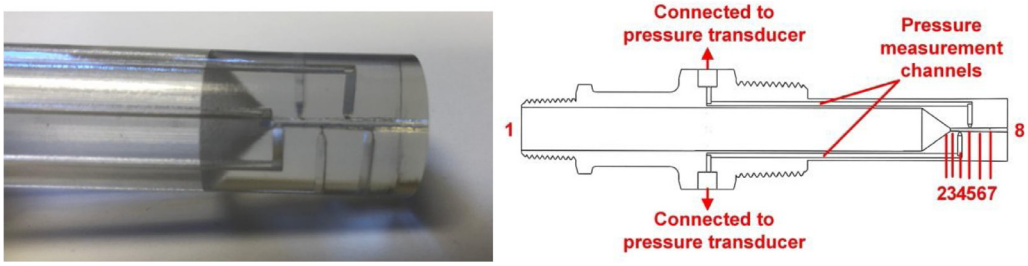


Fig. 2. (a) Transparent convergent–divergent nozzle with pressure measurement channels; (b) cross-sectional view of the nozzle.

Table 2
Axial locations of the measurement channel tips relative to the nozzle throat.

Nozzle #	Pressure measurement channel #	2	3	4	5	6	7
1, 3, 4, 5	Axial location of the tip relative to the	−1.0	0.6	3.4	6.9	10.4	13.9
2	nozzle throat (mm)	−0.6	0.7	7.5	14.5	21.5	28.5

Table 3
Summary of measured and calculated property uncertainties.

	Property	Uncertainty	Instrument measurement range
Measured	Thermocouple reading (°C)	±0.5	−200–350
	Pressure reading (kPa)	±2	0–1724 (gauge)
	Mass flow rate reading (g $s^{-1)}$	±0.1	0–50
Calculated	Vortex strength (dimensionless)	±0.01	–
	Subcooling (°C)	±0.5	–
	Inlet vapor quality (dimensionless)	±0.0045	–

difference between the axial and tangential inlets due to the heat transfer with the ambient is avoided during the tests. The temperature difference between the axial and tangential inlets was remained within 0.5 °C when both tangential and axial inlets were open. The axial inlet temperature was assumed to be the nozzle inlet temperature T_{in} when both inlets were open. When one of the inlets was fully closed, the open inlet temperature was assumed to be T_{in}. The total mass flow rate $\dot{m}_{total}$ and the nozzle axial inlet mass flow rate $\dot{m}_{axial}$ were measured by Coriolis-type mass flow meters. The nozzle's tangential inlet mass flow rate $\dot{m}_{tangential}$ can be calculated by subtracting the nozzle axial inlet mass flow rate from the total mass flow rate. The ratio of the nozzle tangential inlet mass flow rate to the total mass flow rate was adjusted by two valves. The larger the ratio, the larger the vortex strength for the same total mass flow rate. In this paper, the vortex strength is defined as the ratio of the nozzle tangential inlet mass flow rate to the total mass flow rate, which can be expressed as shown in the following equation:

$$VS = \dot{m}_{tangential}/\dot{m}_{total} \tag{1}$$

A summary of measured and calculated property uncertainties is provided in Table 3. The measured property uncertainties (thermocouple, pressure, and mass flow rate readings) are all instrumental uncertainties obtained directly from the instrument specifications. The uncertainties of calculated properties (vortex strength, subcooling, and inlet vapor quality) associated with instrumental uncertainties of measured properties are calculated by uncertainty propagation as shown in the following equation (Taylor and Kuyatt, 1994):

$$U_Y = \sqrt{\sum_i \left(\frac{\partial Y}{\partial X_i}\right)^2 U_{X_i}^2} \tag{2}$$

where U_Y is the uncertainty of calculated property and U_{X_i} is the uncertainty of measured property.

Different nozzle inlet pressures were achieved by adjusting the heating water temperature, pump speed, and inlet control valves. The nozzle outlet pressure can be adjusted by a valve installed downstream of the nozzle. For all tests with subcooled liquid at the nozzle inlet, the sight glass installed at the nozzle inlet and the transparent body of the nozzle allows for visual confirmation that no bubbles are present at the nozzle inlet, which provides ultimate confirmation for inlet subcooling.

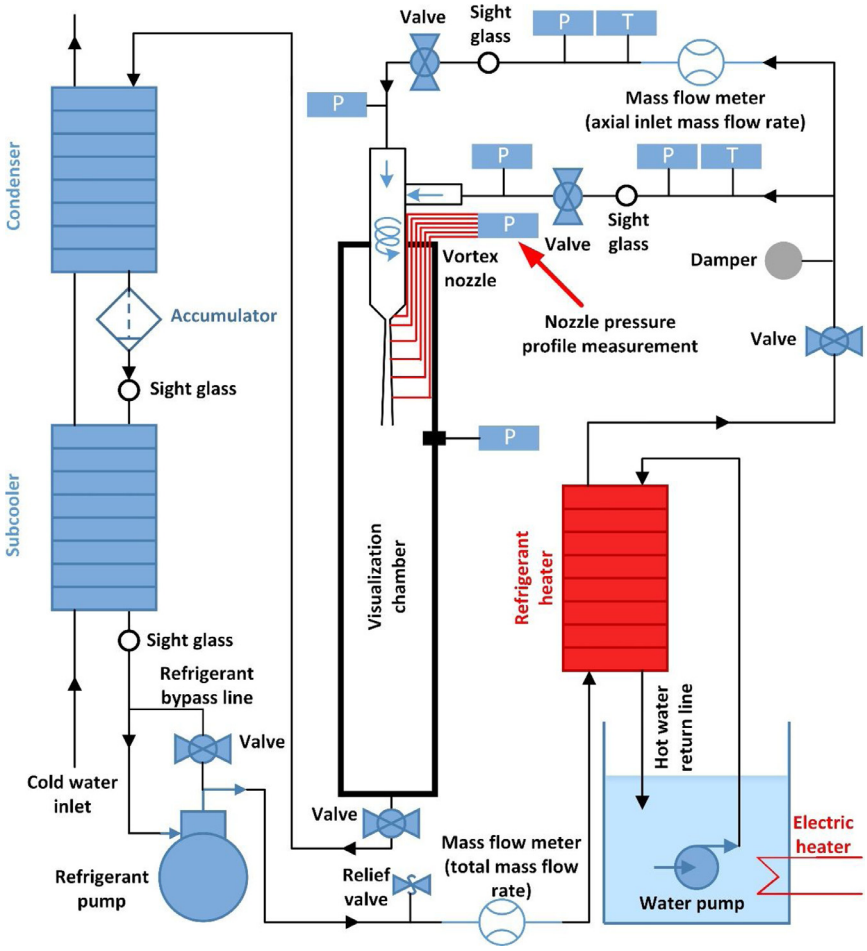


Fig. 3. Experimental facility for measurement of pressure profiles of vortex flashing flows in convergent–divergent nozzles.

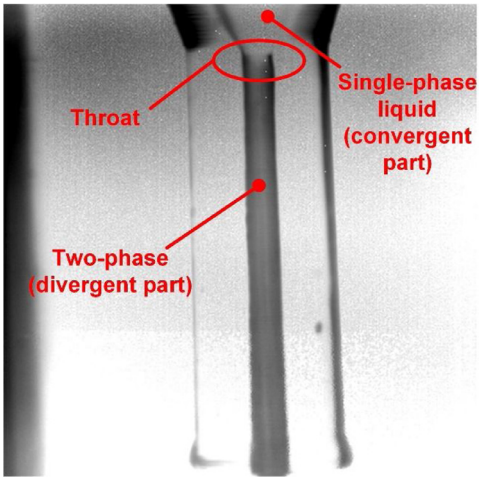


Fig. 4. Visualization of initially subcooled flashing nozzle flow.

3. Estimation of vapor qualities with 1D model

Based on the pressure profiles and mass flow rates obtained through the experiment, the vapor qualities of the initially subcooled flashing flow in the divergent part of the nozzle can be estimated with the following 1D model.

According to the previous visualization results of initially subcooled flashing flow (Zhu and Elbel, 2018) as shown in Fig. 4, it is assumed that the fluid remains single-phase liquid in the convergent part of the nozzle and the vapor quality at location 3 (immediately downstream of the throat) of each tested nozzle is zero. Vapor and liquid velocities are assumed to be equal at each location, which appears to be justifiable as the visualization showed dispersed vapor phase. Pressure and axial velocity are assumed to be uniform at each cross-section. For a segment of the divergent part between locations i and j, flow mass and axial momentum balance equations are shown in the following equations:

$$\rho_i V_i A_i = \rho_j V_j A_j = \dot{m}_{total} \tag{3}$$

$$\sum F + \dot{m}_{total}(V_i - V_j) = 0 \tag{4}$$

where $V_{i|j}$ is the axial flow velocity, $\rho_{i|j}$ is assumed to be the density of thermodynamic equilibrium two-phase fluid at pressure $P_{i|j}$ and vapor quality $x_{i|j}$, $A_{i|j}$ is the flow area.

The forces considered are pressure forces and frictional force between the wall and the fluid. For the estimation of frictional force, two different approaches (homogeneous flow approach and separated flow approach) have been used.

3.1. Homogeneous flow approach

The frictional force is estimated with Swamee-Jain (1976) equation:

$$f = \frac{0.25}{\left[\log\left(\frac{\epsilon}{D}\frac{1}{3.7} + \frac{5.74}{Re^{0.9}}\right)\right]^2} \tag{5}$$

$$\tau_w = \frac{f\rho V^2}{8} \tag{6}$$

where average values of ρ, V, D at locations i and j have been used for estimation of Reynolds number Re and wall shear stress τ_w. For the two-phase flow viscosity, three different models have been considered:

$$\text{Owens}(1961): \mu_{tp} = \mu_l \tag{7}$$

$$\text{McAdams}(1942): \frac{1}{\mu_{tp}} = \frac{x}{\mu_v} + \frac{1-x}{\mu_l} \tag{8}$$

$$\text{Cicchitti } et\ al.\ (1960): \mu_{tp} = x\mu_v + (1-x)\mu_l \tag{9}$$

The sensitivity of the choice of two-phase flow viscosity model will be examined in a later section. ϵ is the surface roughness. The nozzle inner surface roughness was measured with Dektak 3ST surface profilometer. The arithmetical mean roughness (Ra) was measured to be 1.6 μm.

3.2. Separated flow approach

Lockhart and Martinelli (1949) proposed the original correlation scheme for two-phase flows that predicted the two-phase frictional pressure drop based on a friction multiplier for the liquid-phase or the vapor-phase. It was assumed that the static pressure drop of the liquid and gas phases flowing simultaneously is equal at any point along the channel. A few parameters have been introduced:

The ϕ_l^2 parameter is defined as

$$\phi_l^2 = \frac{\left(\frac{dP}{dz}\right)_{tp}}{\left(\frac{dP}{dz}\right)_l} \tag{10}$$

which is the ratio of two-phase pressure gradient due to friction in a channel flow at mass flow rate $\dot{m}_{tp}$ to the pressure gradient for a liquid stream flowing alone in the same channel at mass flow rate $\dot{m}_l = (1-x)\dot{m}_{tp}$.

The parameter R is defined as

$$R = \left[\frac{\left(\frac{dP}{dz}\right)_l}{\left(\frac{dP}{dz}\right)_v}\right]^{\frac{1}{2}} \tag{11}$$

which is the ratio of pressure gradient due to friction of liquid phase flowing at mass flow rate $\dot{m}_l$ to the pressure gradient for vapor phase flowing alone in the same channel at mass flow rate $\dot{m}_v = x\dot{m}_{tp}$.

Chisholm (1967) proposed to relate ϕ_l^2 and R by an analytical correlation:

$$\phi_l^2 = 1 + \frac{C}{R} + \frac{1}{R^2} \tag{12}$$

where C is a constant which depends on whether the vapor and liquid flows are laminar or turbulent. When both the liquid and vapor phases are turbulent, C is taken as 20.

It should be noted that the correlations used in this paper existed for decades and there are a number of newer correlations proposed. New correlations are usually developed for specific flow types. For the flow type investigated in the current paper (vortex flashing flow), to the best of the authors' knowledge, it has not been well studied before and there is no specific correlation available. Moreover, in the current study, the authors have limited capability to evaluate the suitability of correlation. Therefore, less flow type-dependent correlations are used in the current study instead of newer and more flow-type dependent ones.

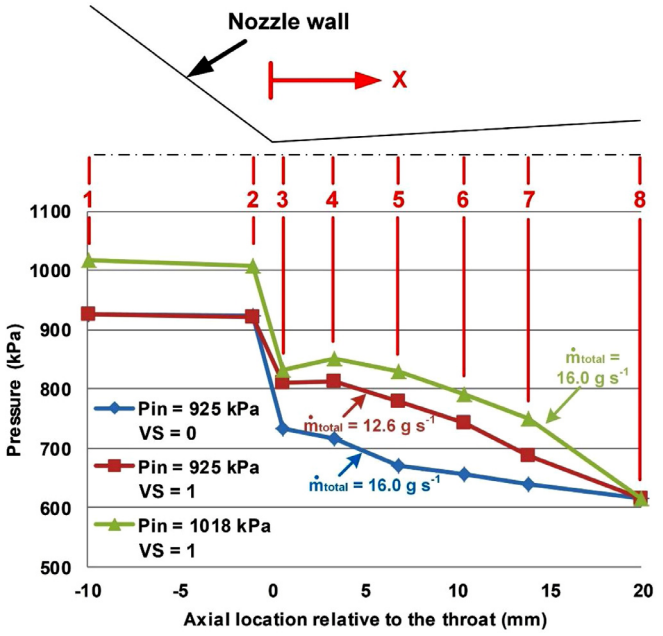


Fig. 5. Pressure at different axial locations of nozzle 1 when $T_{in} = 36.0$ °C and $P_{out} = 615$ kPa.

Since vapor quality at location 3 has been assumed as zero, vapor quality at location 4 can therefore be computed with the above assumptions and model provided the pressure profile and mass flow rate data. The vapor qualities at locations downstream can be computed in a similar fashion.

4. Results and discussion

4.1. Influence of inlet vortex on initially subcooled flashing nozzle flow pressure profile

Nozzle 1 was first tested under the conditions $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, $P_{out} = 615$ kPa with $VS = 0$ and $VS = 1$ and the conditions $P_{in} = 1018$ kPa, $T_{in} = 36.0$ °C, $P_{out} = 615$ kPa with $VS = 1$. The pressure at different axial locations of the nozzle is presented in Fig. 5. With $P_{in} = 925$ kPa (visual confirmation through the sight glass and the transparent body of the nozzle that no bubbles are present at the nozzle inlet), when VS was changed from 0 to 1, the nozzle mass flow rate was reduced significantly from 16.0 g s^{-1} to 12.6 g s^{-1}. Meanwhile, the pressure drop across the divergent part was increased due to the vortex. The pressure difference between locations 3 and 8 is 117 kPa when $VS = 0$ and is increased to 196 kPa when $VS = 1$ (Fig. 6). It is believed that this is due to the increased amount of vapor generated after the application of inlet vortex. It has been observed in previous flow visualization that in the convergent part of the nozzle, the fluid remains single-phase liquid. With elevated pressure at the throat and constant inlet conditions, the pressure difference across the nozzle convergent part decreases and the nozzle mass flow rate is reduced. With higher inlet pressure $P_{in} = 1018$ kPa and maximum inlet vortex, same mass flow rate (16.0 g s^{-1}) as that of the case with $P_{in} = 925$ kPa and no inlet vortex has been achieved. The pressure difference between locations 3 and 8 is increased from 117 kPa to 217 kPa when VS varies from 0 to 1. When $VS = 1$, higher inlet pressure is required to drive the same mass flow rate as when $VS = 0$, which also suggests that the nozzle restrictiveness has been increased by the inlet vortex.

Fig. 7 shows the influence of increasing inlet vortex strength on the pressure at different axial locations of nozzle 1 with $P_{in} = 1035$ kPa, $T_{in} = 40.0$ °C, and $P_{out} = 680$ kPa. It can be observed

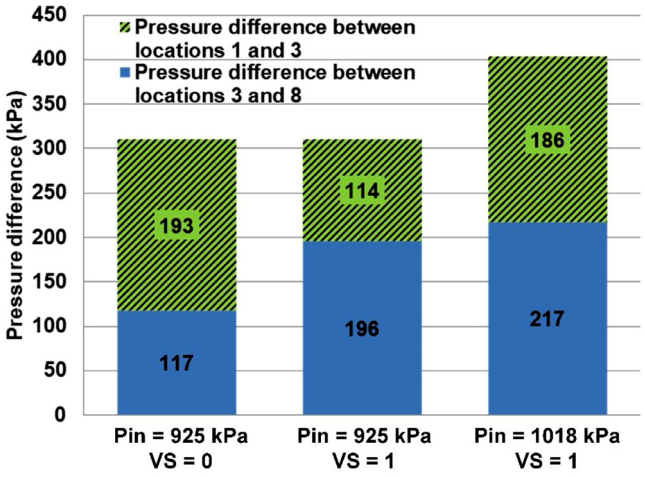


Fig. 6. Pressure differences between axial locations.

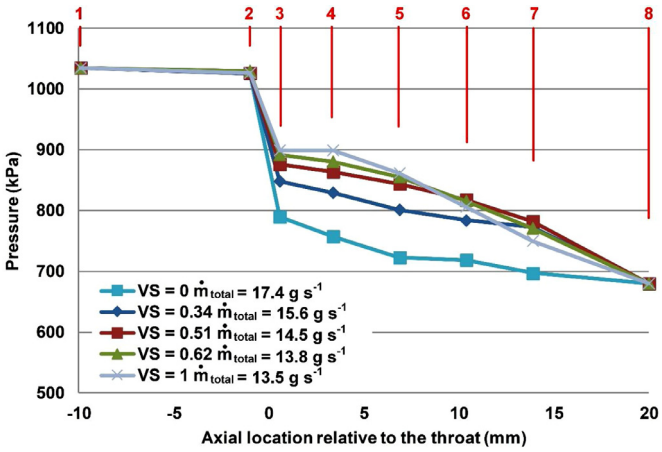


Fig. 7. Influence of inlet vortex strength on the pressure at different axial locations of nozzle 1 with $P_{in} = 1035$ kPa, $T_{in} = 40.0$ °C, and $P_{out} = 680$ kPa.

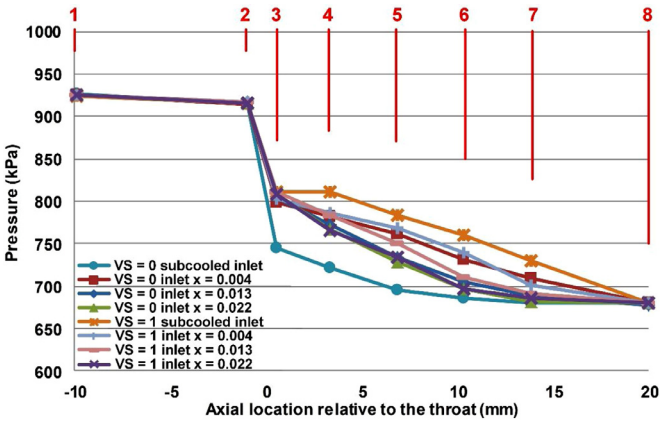


Fig. 8. Influence of inlet vortex on the pressure profile of nozzle 1 with different inlet vapor qualities with $P_{in} = 925$ kPa and $P_{out} = 680$ kPa.

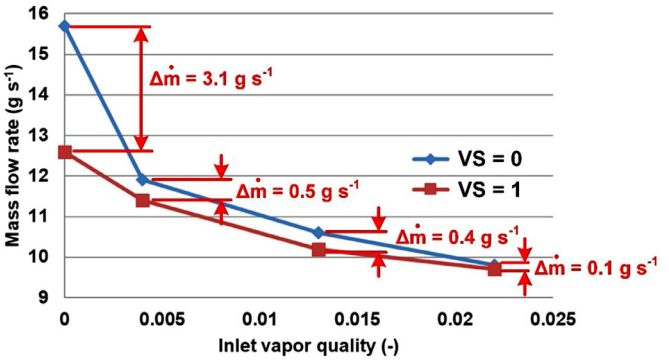


Fig. 9. Influence of inlet vapor quality on the nozzle total mass flow rate with $P_{in} = 925$ kPa and $P_{out} = 680$ kPa.

that as *VS* increases from 0 to 0.34, the pressure drop near the nozzle outlet (between locations 7 and 8) first starts to increase significantly. As the inlet vortex keeps becoming stronger, the increase of pressure drop gradually propagates to the upstream. When *VS* increases from 0 to 0.34, the nozzle mass flow rate is reduced by 1.8 g s^{-1}. However, as *VS* increases from 0.62 to 1, the nozzle mass flow rate is only reduced by 0.3 g s^{-1}. As the vortex strength increases, the pressure drop between locations 3 and 4 near the throat decreases due to the reduced vapor generation caused by elevated pressure and reduced liquid superheat. Therefore, the nozzle throat pressure is relatively unchanged at higher vortex strengths and further increase of vortex strength does not reduce the nozzle mass flow rate significantly.

4.2. Influence of inlet vortex on nozzle flow pressure profile with different inlet vapor qualities

Nozzle 1 was also tested with different inlet vapor qualities. Fig. 8 displays the influence of the inlet vortex on the pressure profile of nozzle 1 with different inlet vapor qualities with $P_{in} = 925$ kPa and $P_{out} = 680$ kPa. For the subcooled inlet cases, $T_{in} = 36.0$ °C and it was visually confirmed through the sight glass and the transparent body of the nozzle that no bubbles were present at the nozzle inlet. Fig. 9 shows the influence of inlet vapor quality on the nozzle total mass flow rate under abovementioned test conditions. For all the cases with two-phase at the nozzle inlet (the existence of two-phase at the nozzle inlet confirmed by visualization through the transparent body of the nozzle), the differences of the pressure profile and the nozzle mass flow rate between the cases $VS = 0$ and 1 (for the same inlet vapor quality) are small, while for the initially subcooled flashing flow the differences are much more significant. It appears that the vortex control effect is achieved by controlling the thermodynamic non-equilibrium (difference between the actual vapor quality and the thermodynamic equilibrium vapor quality) of the fluid. For the cases with two-phase at the inlet, it is hypothesized that due to the availability of large vapor–liquid interfacial area throughout the expansion in both convergent and divergent parts of the nozzle, the fluid remains close to thermodynamic equilibrium. Therefore, in that case, the application of inlet vortex does not effectively increase vapor generation in the nozzle flow due to the small difference between local vapor quality and thermodynamic equilibrium vapor quality and the vortex control effect on nozzle pressure profile and mass flow rate is not significant. The difference between nozzle mass flow rates when $VS = 0$ and 1 keeps decreasing as the inlet vapor quality increases. With an inlet vapor quality of 0.022, the mass flow rate difference is only 0.1 g s^{-1}. For the subcooled inlet cases, the fluid remains single-phase liquid in the convergent part even though the pressure is below saturation pressure. At location 3 near the nozzle throat, the local liquid superheat is 7.3 °C when $VS = 0$. Due to the lack of nucleation sites at the flow center, there is still large thermodynamic non-equilibrium in the divergent part of the nozzle. By applying the vortex, for initially subcooled

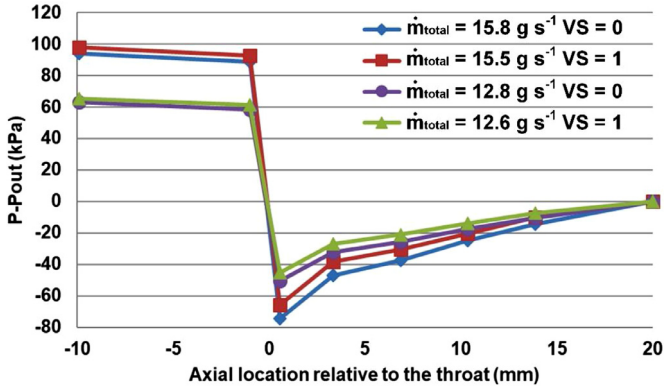


Fig. 10. Pressure relative to the nozzle outlet at different axial locations of nozzle 1 (single-phase liquid throughout the nozzle; liquid temperature: 36.0 °C).

flashing flow, more vapor can be generated from the superheated liquid during the expansion in the nozzle divergent part compared with the case with $VS = 0$, thus both the pressure profile and nozzle mass flow rate are significantly changed.

4.3. Influence of inlet vortex on single-phase liquid nozzle flow pressure profile

In order to investigate the influence of inlet vortex on nozzle flow pressure profile when there is only single-phase liquid throughout the nozzle, nozzle 1 was tested at high pressures. The nozzle inlet temperature was 36.0 °C. Pressures inside the nozzle remained above the corresponding saturation pressure (912 kPa). The pressure relative to the nozzle outlet at different axial locations of nozzle 1 is presented in Fig. 10. It can be observed that the nozzle restrictiveness on single-phase liquid and the pressure profiles under similar conditions do not change significantly as *VS* varies from 0 to 1. The vortex control effect which is shown previously with flashing flow is not seen with single-phase liquid. Moreover, in the divergent part of the nozzle, the pressure rises from the throat to the outlet as the flow area increases and the liquid velocity decreases. In Fig. 5, with initially subcooled flashing flow, pressure drop is observed in the divergent part for similar mass flow rates with $VS = 0$ and 1. This is caused by the vapor generation in the divergent part of the nozzle. It appears that for flashing flow the increase of pressure drop in the divergent part after the application of inlet vortex is mainly due to the increased vapor generation in the divergent part caused by the applied vortex.

4.4. Sensitivity of the choice of frictional force model for vapor quality estimation

Fig. 11 shows the vapor qualities in the divergent part of nozzle 1 at different axial locations relative to the nozzle throat estimated with the 1D models mentioned above using the measured pressure and mass flow rate for the cases $VS = 0$ and 1 at $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C (subcooled nozzle inlet), and $P_{out} = 680$ kPa. The results with homogeneous flow approach using different two-phase flow viscosity models (Owens, 1961; McAdams, 1942; Cicchitti et al.; 1960) as well as separated flow approach using correlation by Lockhart and Martinelli (1949) are presented. The thermodynamic equilibrium vapor qualities assuming isentropic/isenthalpic expansion at each location are also displayed.

It can be observed that the estimated vapor qualities using different two-phase flow viscosity models are close to each other, which is mainly due to the small vapor quality value. In the following sections, only results with Owens (1961) two-phase flow viscosity model (Eq. (7)) will be presented for simplicity. Vapor

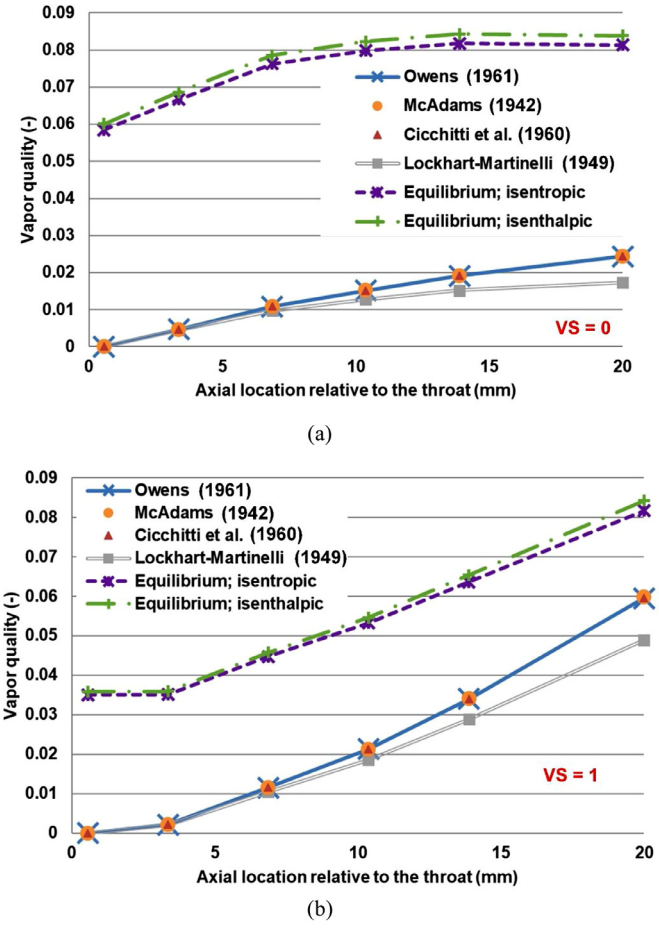


Fig. 11. Estimated vapor qualities and thermodynamic equilibrium vapor qualities assuming isentropic/isenthalpic expansion at different axial locations of nozzle 1 when $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 680$ kPa with (a) $VS = 0$ (b) $VS = 1$.

qualities estimated by Lockhart and Martinelli's (1949) correlation are lower than those by homogeneous flow approach due to the larger predicted frictional force.

Near the throat the estimated vapor qualities for the case with $VS = 1$ are lower than those of the case with $VS = 0$. This is due to the elevated pressure and reduced liquid superheat such that initial vapor generation is suppressed. Between 6.9 mm downstream of the nozzle throat and the outlet, the estimated vapor qualities for the case with $VS = 1$ are higher than those with $VS = 0$. At the nozzle outlet, the vapor quality estimated with Lockhart and Martinelli's (1949) correlation for the case with $VS = 1$ is 0.049 which is close to the thermodynamic equilibrium vapor qualities (0.082 and 0.084 for isentropic and isenthalpic expansion, respectively), while the estimated vapor quality for the case with $VS = 0$ is only 0.017. The dimensionless thermodynamic non-equilibrium can be defined as the ratio of the difference between the equilibrium vapor quality and the local vapor quality to the equilibrium vapor quality, as shown in the following equation:.

$$TNE = \frac{x_e - x}{x_e} \tag{13}$$

The isentropic expansion equilibrium vapor quality is assumed to be the equilibrium vapor quality in Eq. (13) in this study. As *VS* increases from 0 to 1, nozzle outflow *TNE* is reduced from 79.3% to 40.2%. The applied vortex helps to generate more vapor in the divergent part and makes the outflow much closer to thermodynamic equilibrium.

Table 4
Estimated values at the outlet of nozzle 1 when $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 680$ kPa with Lockhart and Martinelli's (1949) correlation.

Vortex strength (dimensionless)	Vapor quality (dimensionless)	Average axial velocity (m s^{-1})	Density (kg m^{-3})	Nozzle isentropic efficiency (%)
0	0.017	14.3	747	22.4
1	0.049	19.4	441	41.3

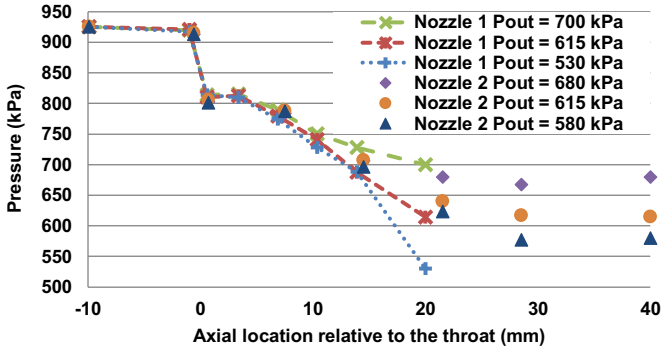


Fig. 12. Pressure at different axial locations of nozzles 1 and 2 with $VS = 1$ when $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C.

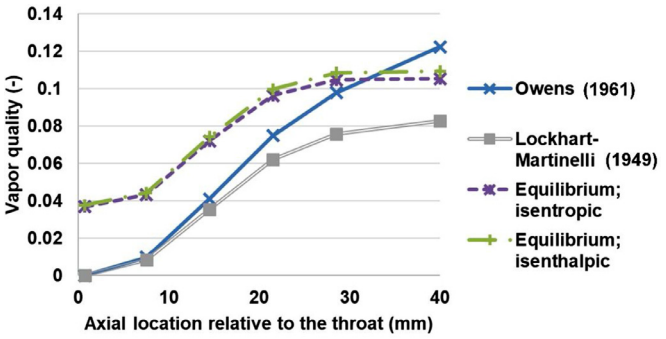


Fig. 13. Estimated vapor qualities in the divergent part of nozzle 2 when $VS = 1$, $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 615$ kPa.

Table 4 shows the estimated values at the outlet of nozzle 1 when $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 680$ kPa with Lockhart and Martinelli's (1949) correlation. The estimated fluid densities at the nozzle outlet for the cases $VS = 0$ and 1 are 747 kg m^{-3} and 441 kg m^{-3}, respectively. Although the total mass flow rate through the nozzle with $VS = 1$ (12.6 g s^{-1}) is much lower than that with $VS = 0$ (15.7 g s^{-1}), the fluid average axial velocity at the outlet with $VS = 1$ (19.4 m s^{-1}) is much higher than that with $VS = 0$ (14.3 m s^{-1}) due to the much lower fluid density at the outlet. For inlet conditions of $P_{in} = 925$ kPa and $T_{in} = 36.0$ °C and outlet pressure of 680 kPa, the isentropic nozzle outflow velocity $V_{isentropic,\,out}$ is 30.2 m s^{-1}. The isentropic efficiency of a nozzle is defined as shown in the following equation:

$$\eta_{isentropic} = \frac{V_{out}^2}{V_{isentropic,out}^2} \tag{14}$$

By the introduction of vortex, the isentropic efficiency of the nozzle has been improved from 22.4% to 41.3%. When the vortex nozzle is used as the motive nozzle in ejectors, the increase of nozzle isentropic efficiency and nozzle outflow velocity for the same operating conditions could be beneficial to ejector performance as more kinetic energy can be transferred to the suction flow and larger pressure lift can be achieved. It should be noted that higher nozzle isentropic efficiency in this case does not necessarily suggest less irreversibility during the expansion inside the nozzle. Higher outflow velocity is mainly achieved by reduced thermodynamic non-equilibrium and increased vapor generation.

4.5. Comparison of nozzles with different divergent part lengths

Nozzle 2 has the same convergent part and divergent angle as nozzle 1, but the divergent part length of nozzle 2 is 40.0 mm which is twice that of nozzle 1. The pressure at different axial locations of nozzle 2 with $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, $VS = 1$, and various outlet pressures is presented in Fig. 12. The pressure profiles of nozzle 1 with the same inlet conditions and similar outlet pressures are also presented for comparison. Flow choking has been observed with both nozzles 1 and 2. The total mass flow rates through nozzle 1 remained at 12.6 g s^{-1} when P_{out} decreased from 700 kPa to 530 kPa, while the total mass flow rates through nozzle 2 remained at 12.2 g s^{-1} when P_{out} decreased from 680 kPa to 580 kPa. The increase of nozzle divergent part length from 20.0 mm to 40.0 mm does not significantly change vortex control characteristics as the choked mass flow rates through the two nozzles under the same inlet conditions when $VS = 1$ are close. For nozzle 2, the flow pressure drop near the nozzle outlet (between 21.5 mm and 40.0 mm downstream of the nozzle throat) is much smaller than in the upstream region between 7.5 mm and 21.5 mm downstream of the nozzle throat.

Fig. 13 shows the estimated vapor qualities in the divergent part of nozzle 2 at different axial locations relative to the nozzle throat when $VS = 1$, $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 615$ kPa. At the nozzle outlet, the estimated vapor quality using Owens (1961) viscosity model is higher than the thermodynamic equilibrium vapor qualities, which suggests that the vapor quality is overestimated and the frictional force is underestimated. The estimated vapor qualities by Lockhart and Martinelli's (1949) correlation are significantly lower than those of Owens (1961) model, especially near the nozzle outlet when the vapor quality is high. They remain lower than the thermodynamic equilibrium vapor qualities. From the nozzle throat to the outlet, vapor generation first increases rapidly and then slows down towards the outlet. As a result, pressure profile near the outlet is much flatter than in the upstream region. The majority of vapor generation takes place in the upstream half of the divergent part. Vapor quality reaches 0.062 at 21.5 mm downstream of the throat and only increases to 0.083 at the outlet (40.0 mm downstream of the throat). The limited vapor generation near the nozzle outlet is due to the reduced thermodynamic non-equilibrium as local vapor quality is already close to the thermodynamic equilibrium vapor qualities.

Table 5 displays the estimated values using Lockhart and Martinelli's (1949) correlation at the outlet of nozzles 1 and 2 when $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 615$ kPa with $VS = 1$. Although higher vapor quality (lower level of thermodynamic non-equilibrium) and lower fluid density can be achieved at the nozzle outlet with longer divergent part length, average axial velocity at the outlet of nozzle 2 is lower than that of nozzle 1. This is mainly due to the larger outlet area as well as the larger frictional losses in the nozzle (larger contact area between the fluid and the nozzle wall). Nozzle isentropic efficiency is reduced from 43.4% to 24.6% when divergent part length is doubled from 20.0 mm to 40.0 mm. If the divergent part length is too short, the time for vapor generation in the nozzle is insufficient and the thermodynamic non-equilibrium in the outflow is still at a high level even with

Table 5
Estimated values at the outlet of nozzles 1 and 2 when $VS = 1$, $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 615$ kPa with Lockhart and Martinelli's (1949) correlation.

Nozzle #	Divergent part length (mm)	Nozzle outlet area (mm^2)	Vapor quality (dimensionless)	Thermodynamic non-equilibrium (%)	Average axial velocity ($m\ s^{-1}$)	Density ($kg\ m^{-3}$)	Nozzle isentropic efficiency (%)
1	20.0	1.47	0.063	40.2	24.6	347	43.4
2	40.0	2.32	0.083	21.2	18.5	284	24.6

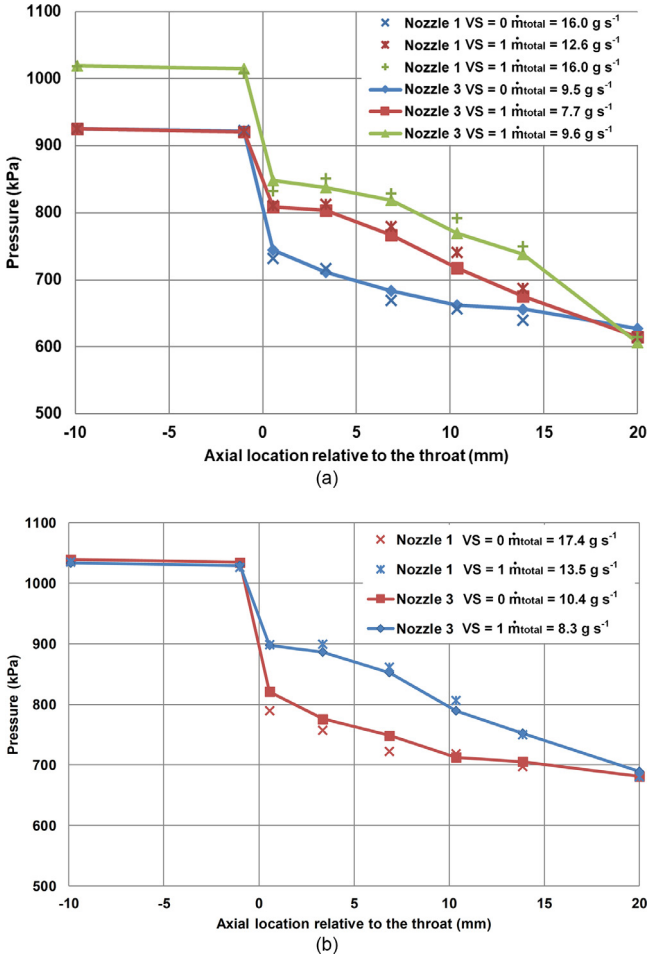


Fig. 14. Comparison of pressure profiles of nozzles 1 and 3 under similar inlet and outlet pressures when (a) $T_{in} = 36.0$ °C (b) $T_{in} = 40.0$ °C.

vortex. This might result in small vortex control range (Zhu and Elbel, 2018) as well as low nozzle isentropic efficiency. Therefore, the nozzle divergent part length needs to be appropriately sized in order to achieve satisfactory vortex nozzle performance.

4.6. Comparison of nozzles with different throat diameters

Nozzle 3 has smaller throat diameter (0.83 mm) than that of nozzle 1 (1.02 mm) but the same convergent part length as well as divergent part length and angle. Fig. 14 shows a comparison of pressure profiles of nozzles 1 and 3 under similar inlet and outlet conditions.

It can be observed that for the same test conditions the pressures at each axial location of nozzles 1 and 3 are close to each other (the differences between the pressure profiles of nozzles 1 and 3 are observed to be less than 29 kPa), while there are significant differences in the mass flow rates through nozzles 1 and 3 due to the change in flow area. The throat areas of nozzles 1 and 3 are 0.82 mm^2 and 0.54 mm^2, respectively. When

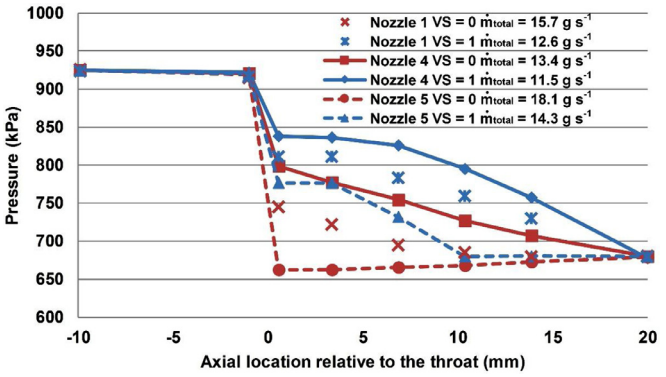
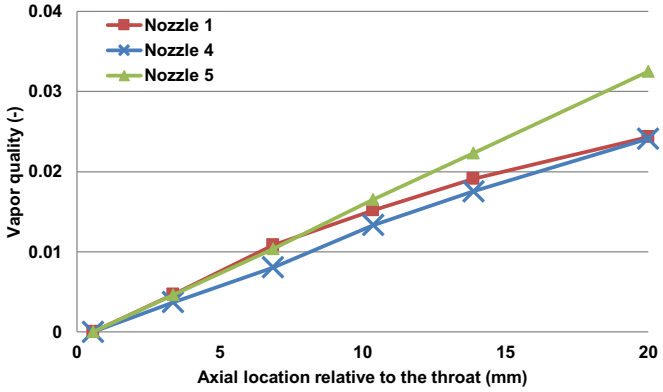


Fig. 15. Comparison of pressure profiles of nozzles 1, 4 and 5 when $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, $P_{out} = 680$ kPa.

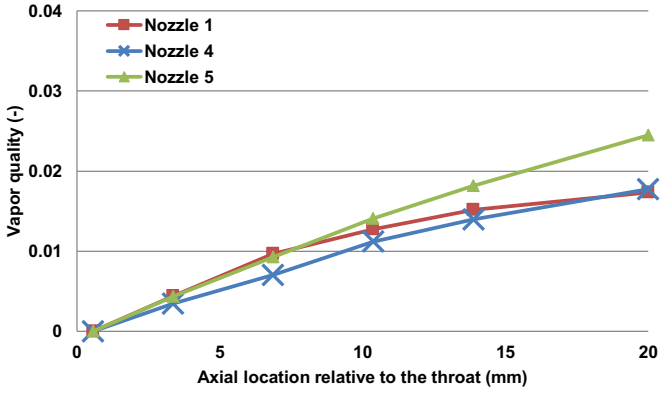
$VS = 0$, $P_{in} = 1035$ kPa, $T_{in} = 40.0$ °C, $P_{out} = 680$ kPa, the mass flow rate going through nozzle 1 was 17.4 g s^{-1}. Under the same inlet and outlet temperature and pressure conditions with $VS = 1$, the mass flow rate through nozzle 1 was reduced to 13.5 g s^{-1}. For nozzle 3 when $VS = 0$, $P_{in} = 1035$ kPa, $T_{in} = 40.0$ °C, $P_{out} = 681$ kPa, the mass flow rate was 10.4 g s^{-1}. By linear interpolation between (0.82 mm^2, 17.4 g s^{-1}) and (0.54 mm^2, 10.4 g s^{-1}), the nozzle throat area for which the nozzle mass flow rate is 13.5 g s^{-1} when $VS = 0$, $P_{in} = 1035$ kPa, $T_{in} = 40.0$ °C, $P_{out} = 680$ kPa (13.5 g s^{-1} was the mass flow rate through nozzle 1 under the same conditions with maximum vortex applied) is estimated to be 0.66 mm^2. The corresponding nozzle throat diameter is 0.92 mm. Therefore, with nozzle inlet vortex, the same nozzle restrictiveness can be achieved with 24% larger throat area and 11% larger throat diameter, which suggests that applying nozzle inlet vortex can allow particles with larger size to pass through and help to reduce the nozzle vulnerability to clogging for equivalent nozzle restrictiveness.

4.7. Comparison of nozzles with different divergent angles

Nozzles 1, 4, and 5 have the same convergent part, throat diameter, and divergent part length but different divergent angles (1.0°, 0.5°, and 2.0° full divergent angle, respectively). Fig. 15 compares the pressure profiles of nozzles 1, 4, and 5 under similar inlet and outlet conditions. Figs. 16 and 17 display the estimated vapor qualities in the divergent part of nozzles 1, 4, and 5 when $VS = 0$ and 1, respectively. When $VS = 0$, nozzle 4 with the smallest divergent angle among the three nozzles (therefore the smallest flow area at each axial location) has the largest pressure drop across the divergent part, even though the vapor quality at each axial location is similar and nozzle mass flow rate is the lowest. When $VS = 1$, more vapor is generated in nozzle 5 with the largest divergent angle, especially at the immediate downstream of the nozzle throat. Vapor generation near the throat is less suppressed in nozzle 5 due to the lower pressure level and therefore larger liquid superheat. The relative vortex control range is defined as the ratio of the difference in nozzle total mass flow rates when $VS = 0$ and 1 to the nozzle mass flow rate when $VS = 0$. Nozzle 4 with the smallest divergent angle has lower vortex control range (14%) than nozzles 1 and 5

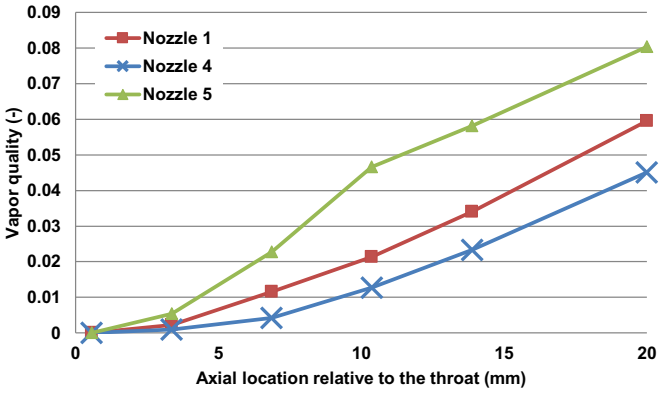


(a)

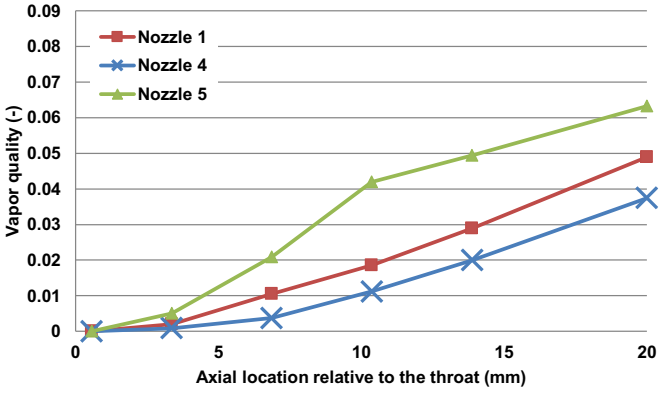


(b)

Fig. 16. Estimated vapor qualities in the divergent part of nozzles 1, 4, and 5 when $VS = 0$, $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 680$ kPa using (a) Owens (1961) viscosity model (b) Lockhart and Martinelli's (1949) correlation.



(a)



(b)

Fig. 17. Estimated vapor qualities in the divergent part of nozzles 1, 4, and 5 when $VS = 1$, $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 680$ kPa using (a) Owens (1961) viscosity model (b) Lockhart and Martinelli's (1949) correlation.

(20% and 21%, respectively) as the vapor generation and pressure drop near the throat are more suppressed when $VS = 1$. When the divergent angle is too large, two-phase nozzle flow can be separated from the nozzle wall and recirculation of ambient vapor can take place between the nozzle wall and the two-phase nozzle flow, as is shown in Zhu and Elbel (2018) by flow visualization. At the extreme case when the full divergent angle is 180° (nozzle divergent part wall is perpendicular to the axial direction of the nozzle flow), the divergent part nozzle wall has no influence on the nozzle flow and the nozzle is equivalent to a convergent-only nozzle. The vortex control range drops to an insignificant level in this case.

Fig. 18 shows the estimated nozzle outlet average axial velocity of nozzles 1, 4, and 5 with different divergent angles using Lockhart and Martinelli's (1949) correlation when $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 680$ kPa. It can be seen that under the same conditions as nozzle divergent angle decreases, nozzle outlet average axial velocity increases due to the reduced nozzle outlet area. When the nozzle is used as motive nozzle in an ejector, higher nozzle outflow velocity under the same conditions could be beneficial to the ejector performance. Choosing the appropriate nozzle divergent angle is needed for vortex nozzle used in ejectors such that satisfactory vortex control range and high ejector performance can be achieved.

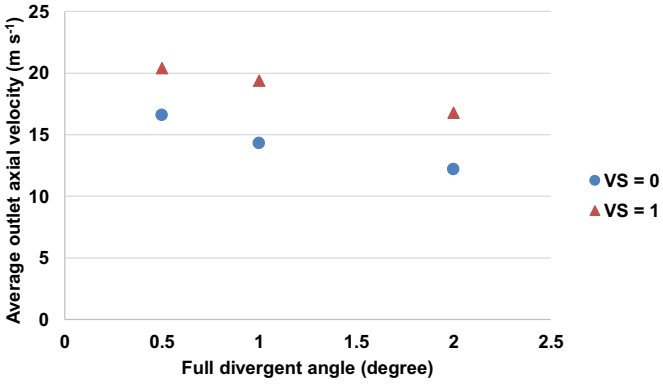


Fig. 18. Estimated nozzle outlet average axial velocity of nozzles 1, 4, and 5 with different divergent angles using Lockhart and Martinelli's (1949) correlation when $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, and $P_{out} = 680$ kPa.

4.8. Nozzle outflow visualization

Fig. 19 shows the outflows of nozzle 4 with $VS = 0$ and $VS = 1$ when $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, $P_{out} = 700$ kPa. It can be observed that the application of inlet vortex has increased the expansion angle (angle between boundary line of the outflow and the centerline) of nozzle outflow from 0.2° to 3.0°. Despite the dissipation during the expansion through the convergent–divergent nozzle, when $VS = 1$ the tangential velocity of the nozzle outflow is still non-zero, which appears to be the major contributor to the increased expansion angle. When a vortex nozzle is used as the motive nozzle in an ejector, the increase of expansion angle might have impact on ejector performance (Zhu et al., 2017). The motive flow and the suction flow can be obstructed at the mixing section entrance due to the large expansion angle which reduces the

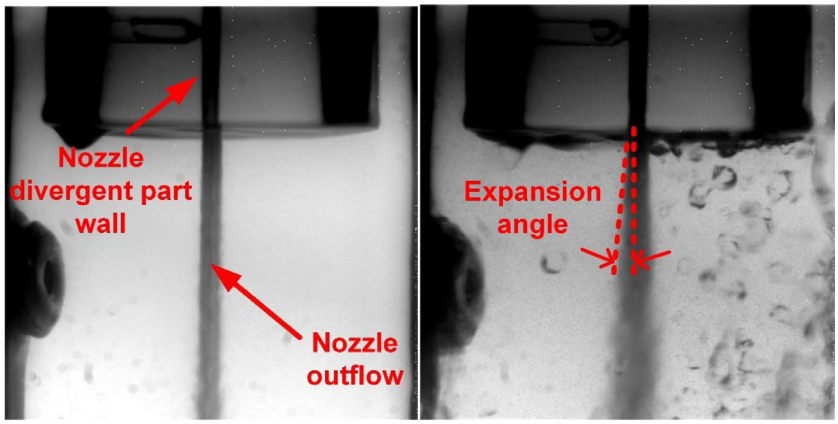


Fig. 19. Comparison of nozzle 4 outflows with (a) $VS = 0$ (b) $VS = 1$ when $P_{in} = 925$ kPa, $T_{in} = 36.0$ °C, $P_{out} = 700$ kPa.

entrainment performance. This should be taken into consideration during the design of ejector with vortex nozzle.

5. Conclusions

In this study, pressure profiles of vortex flashing R134a flows expanded through convergent–divergent nozzles under various conditions have been measured. A 1D model for the estimation of vapor qualities in the initially subcooled flashing nozzle flow based on the measurement results was also developed. The following results were obtained:

1. After the introduction of an inlet vortex to the initially subcooled flashing nozzle flow, the pressure drop across the divergent part has been increased. As observed by flow visualization, the fluid is single-phase liquid in the convergent part of the nozzle. For constant inlet and outlet conditions, with elevated pressure at the nozzle throat after the inlet vortex is applied, the nozzle convergent part pressure difference is reduced and the nozzle mass flow rate decreases. To drive the same mass flow rate with inlet vortex applied, higher inlet pressure is required.
2. When the inlet vortex is applied to flashing nozzle flow with two-phase at the inlet, the differences of the pressure profile and the nozzle mass flow rate between the cases $VS = 0$ and 1 are small, while for initially subcooled flashing flow the differences are much more significant. It appears that the vortex control effect is achieved by controlling the thermodynamic non-equilibrium of the fluid. For the cases with two-phase at the inlet, due to the availability of large vapor–liquid interfacial area throughout the expansion in both convergent and divergent parts of the nozzle, the fluid is believed to be close to the thermodynamic equilibrium. Therefore, in that case, the application of inlet vortex does not effectively increase vapor generation in the nozzle flow due to the small difference between local vapor quality and thermodynamic equilibrium vapor quality. Consequently, the vortex control effect on nozzle pressure profile and mass flow rate is not significant.
3. When the inlet vortex is applied to single-phase liquid nozzle flow, the nozzle restrictiveness on single-phase liquid and the pressure profiles under similar conditions do not change significantly. Moreover, in the divergent part of the nozzle, pressure rises from the throat to the outlet as the flow area increases and the liquid velocity decreases.
4. Using the 1D model presented in this paper, it was shown that for initially subcooled flashing nozzle flow under the same inlet and outlet conditions, the applied vortex helps to generate more vapor in the divergent part and makes the outflow much closer to thermodynamic equilibrium. This is why pressure drop across the divergent part has been increased with the vortex. Near the nozzle throat the estimated vapor qualities for the case with $VS = 1$ can be lower than those of the case with $VS = 0$. This is due to the elevated pressure and reduced liquid superheat such that initial vapor generation is suppressed.
5. By the introduction of vortex, the nozzle outflow velocity and the isentropic efficiency of the nozzle can be significantly increased, which could be beneficial to ejector performance when the vortex nozzle is used in ejectors. Higher outflow velocity is mainly achieved by reduced thermodynamic non-equilibrium and increased vapor generation.
6. The nozzle divergent part length needs to be appropriately sized in order to achieve sufficient vortex control range, high nozzle outflow velocity and high nozzle isentropic efficiency. If the length is too short, the time for vapor generation in the nozzle is insufficient and the thermodynamic non-equilibrium in the outflow is still at a high level. If the length is too long, the outflow velocity becomes low due to the large nozzle outlet area and the large frictional losses.
7. Small divergent angles result in large pressure drop across the divergent part due to small flow area. Nozzles with smaller divergent angle might have smaller vortex control range because of the suppressed vapor generation and pressure drop near the throat when $VS = 1$. Under the same conditions as nozzle divergent angle decreases, nozzle outlet average axial velocity increases due to the reduced nozzle outlet area. Choosing the appropriate nozzle divergent angle is needed for vortex nozzle used in ejectors such that satisfactory vortex control range and high ejector performance can be achieved.
8. The application of inlet vortex has increased the expansion angle of the nozzle outflow due to the non-zero tangential velocity of the nozzle outflow. When a vortex nozzle is used as the motive nozzle in an ejector, the increase of expansion angle might result in obstruction of motive flow and suction flow at the mixing section entrance and could have negative impact on ejector performance, which needs to be considered during the design of the ejector.

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